OENG1209: Engineering Science

Week 4:

Rotational Motion

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Today's Topics

1. Kinematics



2. Dynamics



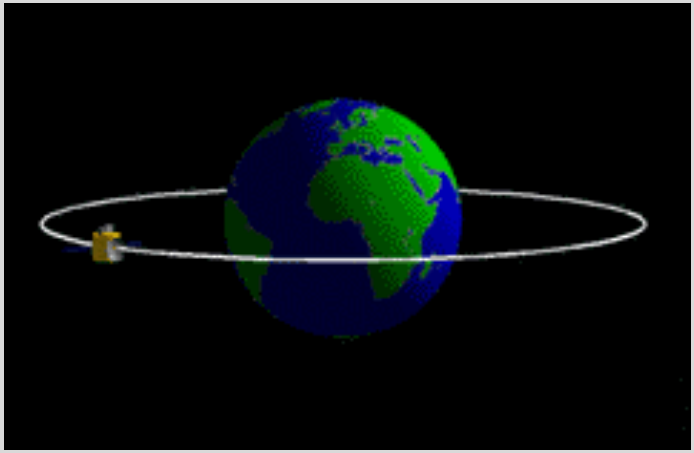
1. Kinematics

Round and round they go

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Motivation

- Consider the motion of a satellite in orbit around the Earth.
- How do we determine its **position**, **velocity** and **acceleration**?



General kinematic relations

Kinematics & Dynamics

Kinematic

Position → Velocity → Acceleration ∝ Force

dynamics

- Position of a particle: $\vec{r}$ vector quantity
- Velocity

$$\vec{v} = \frac{d\vec{r}}{dt}$$

- Acceleration

$$\vec{a} = \frac{d\vec{v}}{dt}$$

General kinematic relations

Direction of velocity

$$\vec{v} = \frac{d\vec{r}}{dt} = \lim_{\Delta t \to 0} \frac{\Delta \vec{r}}{\Delta t}$$ when $\Delta t = 0$ → angle → 90°

is aways perpendicular to $\vec{r}$.

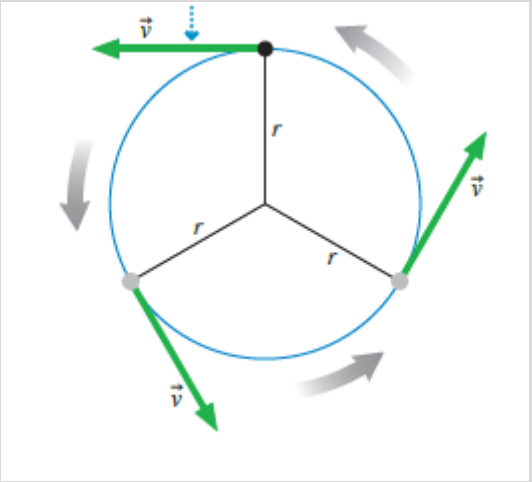
Proof:

Circular motion

The **magniture** of position is constant

$$|\vec{r}] = r = const$$

direction → change
magnitude is NOT changed



Two types:

- **Uniform** (speed is contant)

  acceleration mag → constant
  direction change

- **Nonuniform** (speed changes)

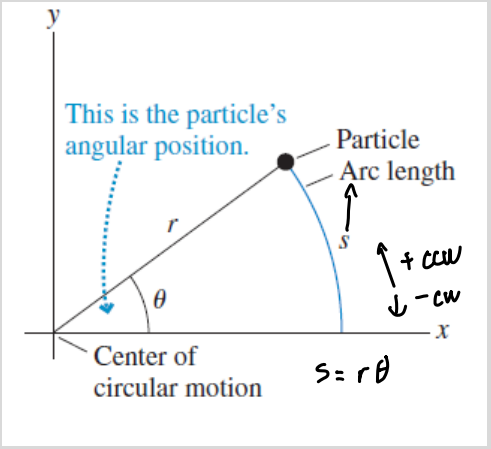
  2 acceleration
  $\vec{v}$ < magnitude → IF changed → acceleration
  direction → IF changed → acceleration

Circular motion
Angular position

Cartesian coordinates (x, y)→ Polar coordinates (r, θ): **Sign convention**

Arc-length coordinate is defined based the angle θ in radians such that

$$s = r\theta$$



Converting radians and degrees

$\theta_{\text{full circle}} = \frac{s}{r} = \cdots$ $360^0 = \frac{\text{Circumference}}{r} = \frac{2\pi r}{r} = 2\pi$

So $1 \text{ rev} = 360^0 = 2\pi \text{ rad}$

1 revolution = 1 round

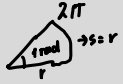
Example: converting an angle of 1 rad to degrees

Soln:

2π rad → 360^0

1 rad → ?

$\frac{360}{2\pi} = 57.3^0$

1 rad → $\frac{s}{r}$ ∴ 

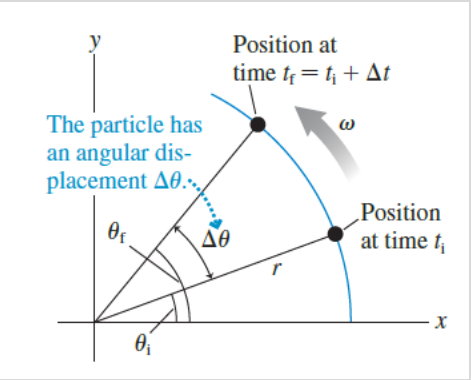
Cicular motion
Angular Velocity

derivative of product
$(uv)' = u'v + uv'$

Computing velocity from arc-length coordinate:

$$s = r\theta \Rightarrow v = \frac{ds}{dt} = \cdots = r\frac{d\theta}{dt}$$

Proof: $v = \frac{d(r\theta)}{dt} = r\frac{d\theta}{dt} + \theta\frac{dr}{dt} = r\cdot\omega$ (r is constant, so $\frac{dr}{dt} = 0$)



Define angular velocity as

$$\omega = \frac{d\theta}{dt}$$

at $t_i \to \theta_i$, $t_f \to \theta_f$ $\Big\}$ $\Delta\theta = \theta_f - \theta_i$

$\omega = \frac{\Delta\theta}{\Delta t}$ → avg angular velocity

$\lim_{t\to 0} \frac{d\theta}{dt}$ → instaneous angular velocity

So the velocity

$$v = r\omega$$

Circular motion
Angular velocity

Example (computing velocity): A 4-cm-diameter crankshaft turns at 2400 rpm (revolutions per minute). What is the speed of a point on the surface of the crankshaft? and the angular velocity?



Soln:

$V, \omega = ?$

$$V = s/\Delta t = \frac{2400 \cdot C}{\Delta t} = \frac{2400 \cdot 2\pi\, d/2}{\Delta t} = \frac{2400 \cdot \pi \cdot 0.04\,m}{60\,s} = 5.03\ m/s$$

(m/s)

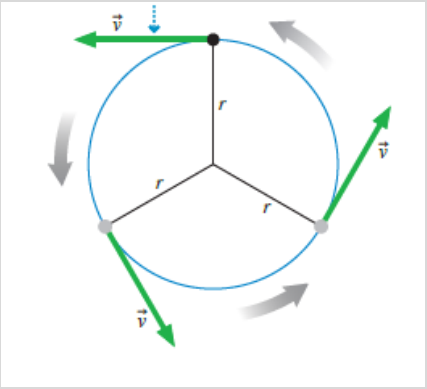
$$V = r\omega$$

$$\omega = V/r = \frac{5.03}{d/2} = \frac{5.03}{0.02\,m} = 251.5\ rad/s$$

Uniform circular motion
Definition

The speed is constant

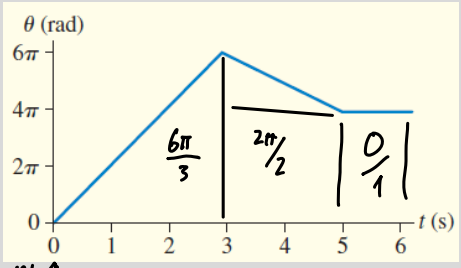
$$v = const$$

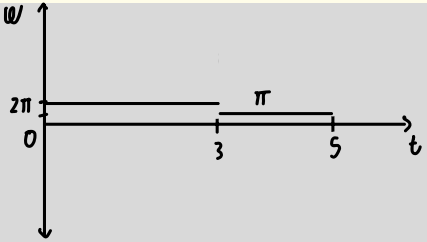


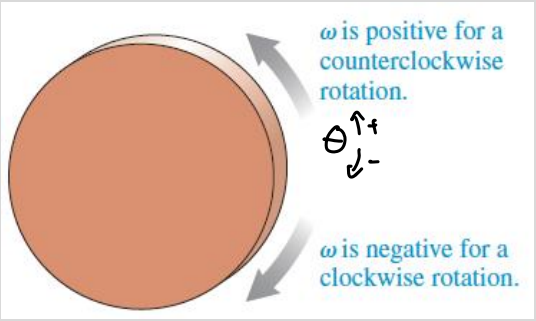
$v = \omega r$ → The angular velocity $\omega = const$

$\omega = d\theta/dt$ → The angular position θ is ? Linear

Example: Given a graph $\theta(t)$, draw $\omega(t)$?







Uniform circular motion

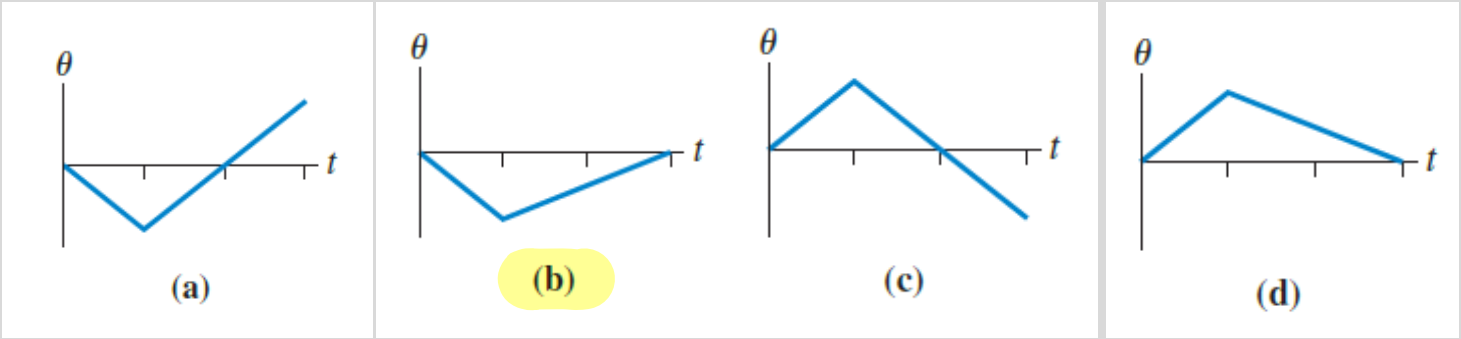
Example

Example: A small steel roulette ball rolls ccw around the inside of a 30 cm diameter roulette wheel. The ball completes 2.0 rev in 1.20 seconds. (a) What is the ball's angular velocity? (b) What is the ball's position at $t = 2.0\ s$? (Assume $\theta_i = 0$)



Uniform cicular motion

Stop to think: A particle moves clockwise around a circle at constant speed for 2.0 s. It then reverses direction and moves counter-clockwise at half the original speed until it has traveled through the same angle. Which one is the particle's angle-versus-time graph?



Ans:

Uniform circular motion

Centripetal acceleration

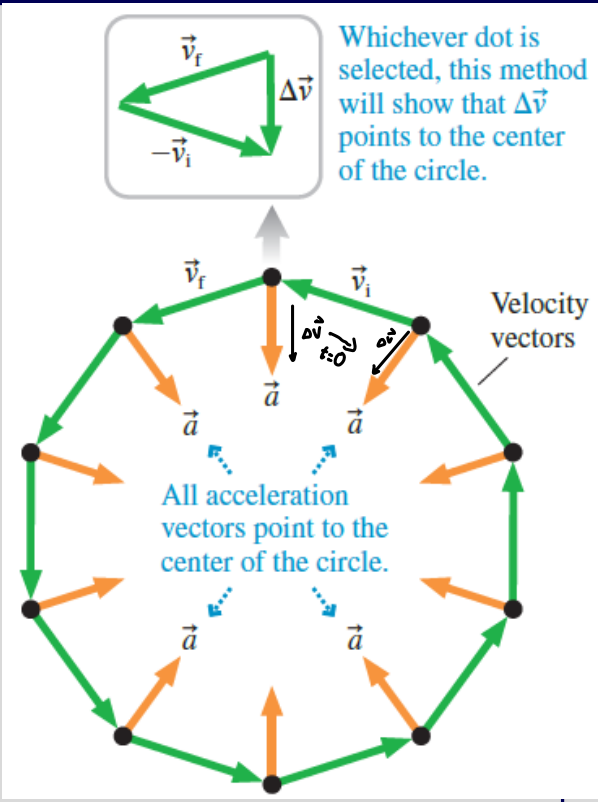
Definition of acceleration

$$\vec{a} = \frac{d\vec{v}}{dt} = \lim_{\Delta t \to 0}\left(\frac{\Delta \vec{v}}{\Delta t}\right) = \lim_{\Delta t \to 0}\left(\frac{\vec{v}_f - \vec{v}_i}{\Delta t}\right)$$

Any observation on the direction of $\vec{a}$?

→ **Centripetal acceleration**

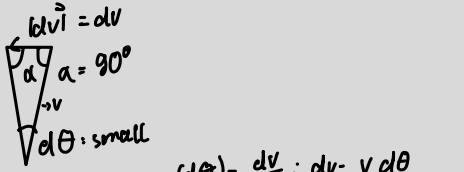
center -> seeking



Uniform circular motion

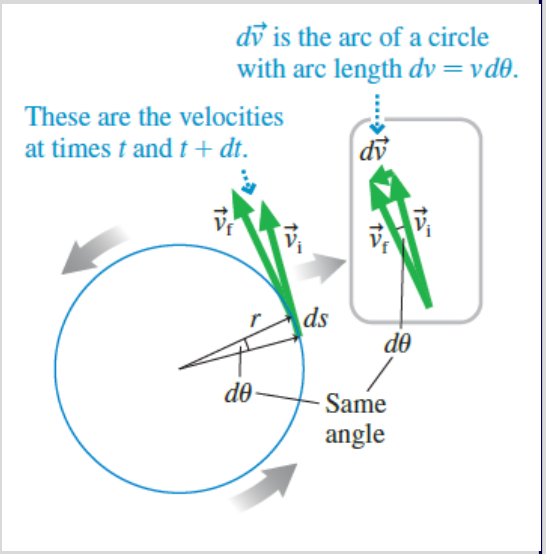
Centripetal acceleration

That is about its direction, what about its **magnitude?**



$d\theta = \approx \tan(d\theta) = \frac{dv}{v} \therefore dv = v\,d\theta$

$a_c = \frac{dv}{dt} = \frac{v\,d\theta}{dt} = v\omega = v \cdot \frac{v}{r} = v^2/r$

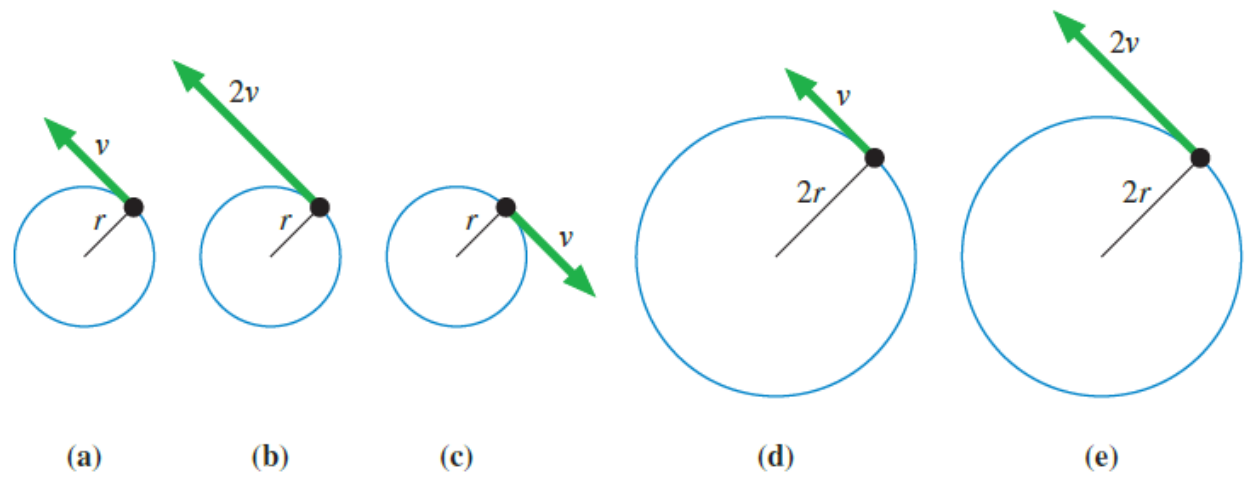


$$a_c = \frac{v^2}{r}$$

Uniform circular motion

Centripetal acceleration

Stop to think: rank in order, from largest to smallest, the centripetal accelerations of particles (a) to (e)?



Ans:

Uniform circular motion
Centripetal acceleration

Example: A typical Ferris Wheel has a radius of 9.0 m and rotates 4.0 times per minute. What speed and acceleration do the riders experience?

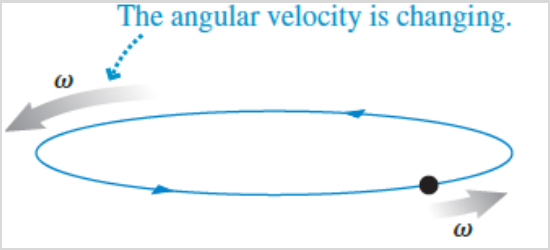


Nonuniform circular motion

Definition

The speed changes

$v \neq const$



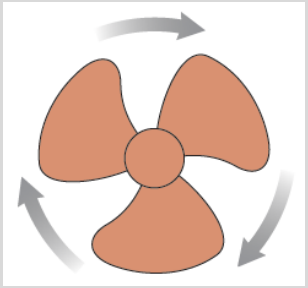
$v = \omega r$ → The angular velocity $\omega \neq const$

We define **the rate of change** of the angular velocity

angular acceleration

$$\alpha = \frac{d\omega}{dt} \rightarrow$$

Stop to think: The fan blade is slowing down. What are the signs of ω and α?



CW → -θ

$w = \frac{d\theta}{dt} < 0$

a. ω is positive and α is positive.

b. ω is positive and α is negative.

c. ω is negative and α is positive.

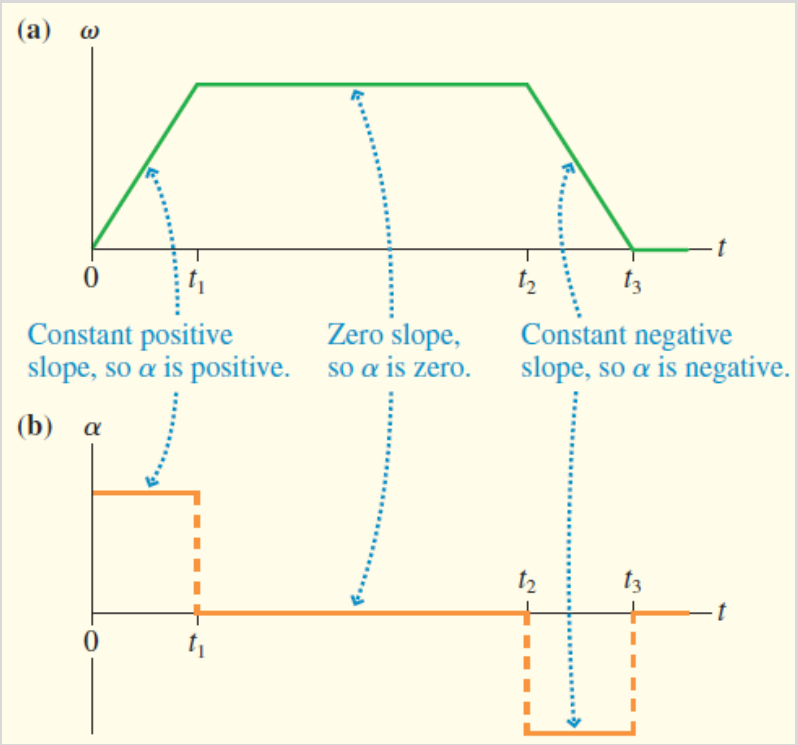
d. ω is negative and α is negative.

Nonuniform circular motion

Definition

$$\alpha = \frac{d\omega}{dt}$$

Graph of $\omega(t)$ → Graph of $\alpha(t)$



Nonuniform circular motion

Definition

The angular acceleration α is constant.

Recall three kinematic equations for 1D motion

$$v = v_0 + a\Delta t; \Delta x = v_0\Delta t + \frac{1}{2}a\Delta t^2; v^2 = v_0^2 + 2a\Delta x$$

Realizing the analogus

$x \leftrightarrow \theta; \quad v \leftrightarrow \omega; \quad a \leftrightarrow \alpha$

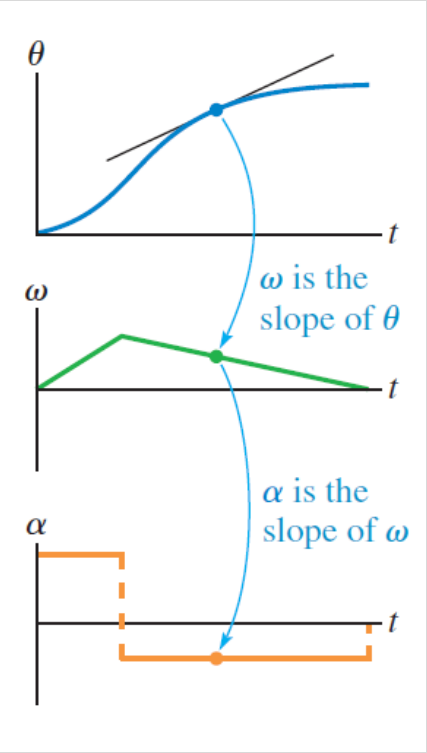
So, three kinematic equations for rotational motion

$\omega = \omega_0 + \alpha\Delta t$

$\Delta\theta = \omega_0\Delta t + \frac{1}{2}\alpha\Delta t^2$

$\omega^2 = \omega_0^2 + 2\alpha\Delta\theta$

This course involve around the idea of Pos, Velo, Acc



Nonuniform circular motion

Example

Example: A ceiling fan spinning at 60 rpm coasts to a stop 25 s after being turned off. How many revolutions does it make while stopping?

Soln:



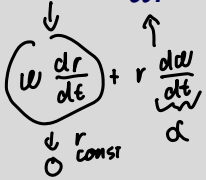
Nonuniform circular motion

Tangent acceleration

Speed changes →

$$a = \frac{dv}{dt} = \frac{d(\omega r)}{dt} = \cdots = \alpha r$$

Proof:



Nonuniform circular motion

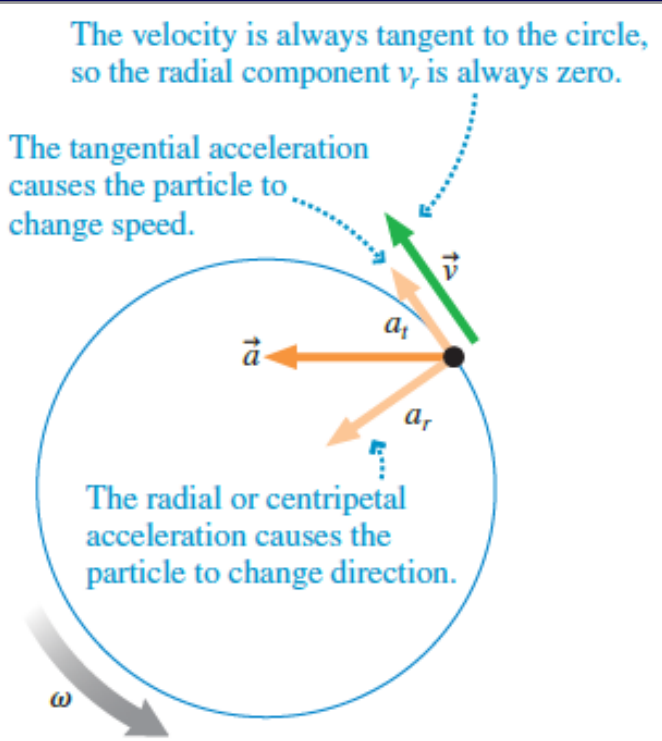
Total acceleration

Total acceleration is computed as

$$a_{total} = \sqrt{a_C^2 + a_T^2}$$

Remarks:

- Velocity is always perpendicular to the radius.
- **Direction** of velocity changes → **centripetal** acceleration.
- **Magnitude** of velocity changes → **tangent** acceleration .



2. Dynamics

How forces cause circular motion

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Uniform circular motion

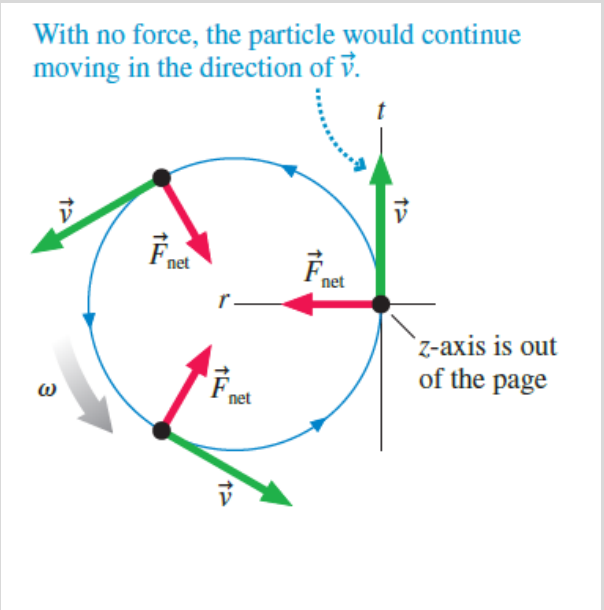
- Recall the definition of **the centripetal acceleration**:

$$|\vec{a}_c| = \frac{v^2}{r} = \omega^2 r$$

- Thus, we can invoke Newton's Second Law to quantify this "centripetal" net force:

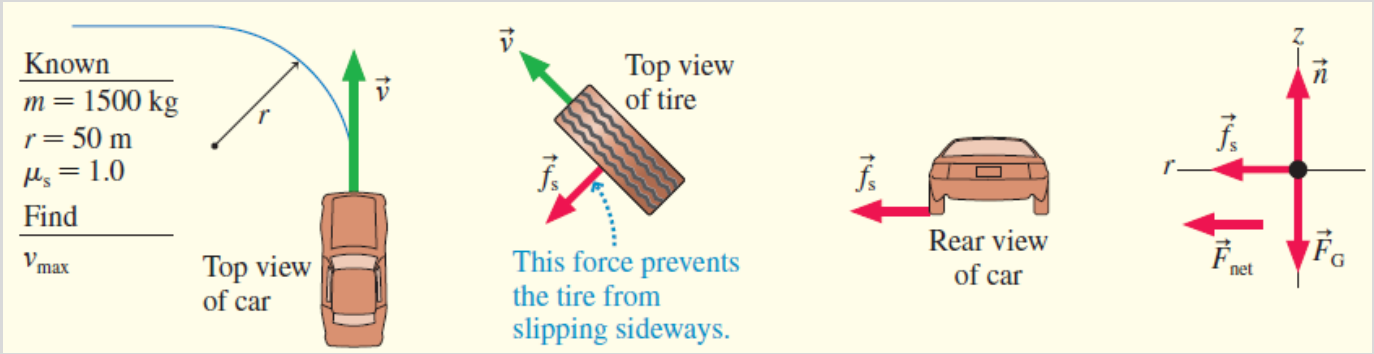
$$\vec{F}_{net} = m\vec{a} = m\frac{v^2}{r}$$

$$(toward\ the\ center\ of\ the\ circle)$$



Uniform circular motion

Example: What is the maximum speed with which a 1500 kg car can make a left turn around a curve of radius 50 m on a level (unbanked) road without sliding? The coefficient of static friction between the tire and the road is $\mu_S = 1.0$



Uniform circular motion

Why does the car stay on track?

Consider a roller coaster going through a loop-the-loop, **why don't the car fall to the ground?**



Uniform circular motion

Why does the car stay on track?

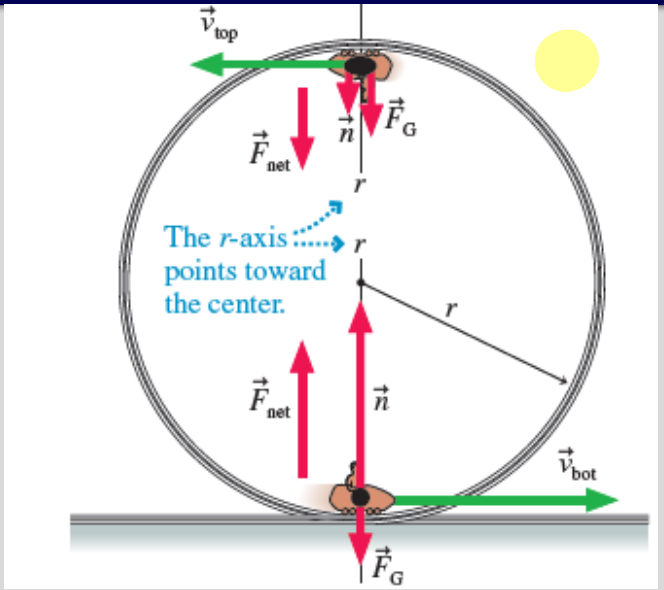
- Newton's 2nd law in the radial direction when the car is **at the top of the loop**:

$$\Sigma F_r = m\,a_r = N_{top} + mg = \frac{mv^2}{r}$$

$$\rightarrow N_{top} = \frac{mv^2}{r} - mg$$

N → should be ⊕

- If N_{top} is zero, there is no contact between the car and the track and the car falls down → **the critical speed** v_C



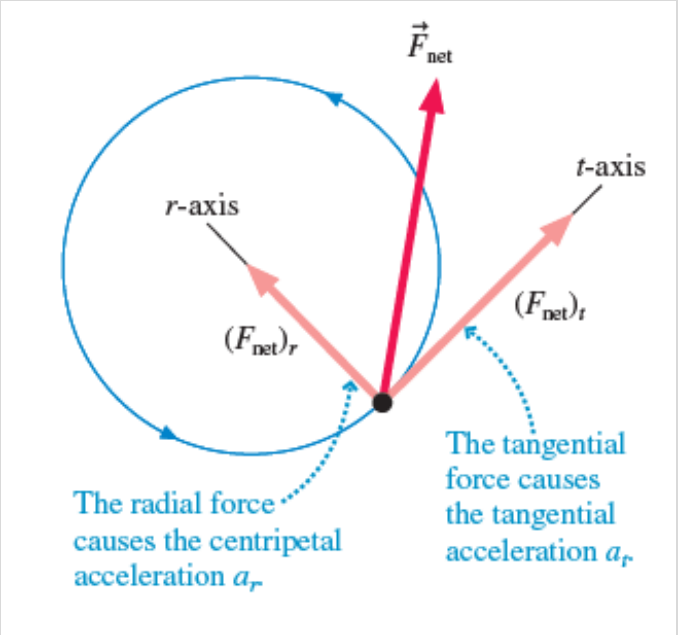
Nonunform circular motion

Net force

Newton's second law

$(F_{net})_r = ma_c = m\frac{v^2}{r}$

$(F_{net})_T = ma_T = m\alpha r$



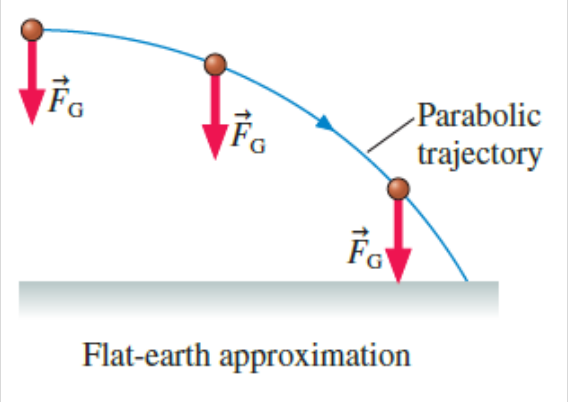
3. An application

Orbit of satellites

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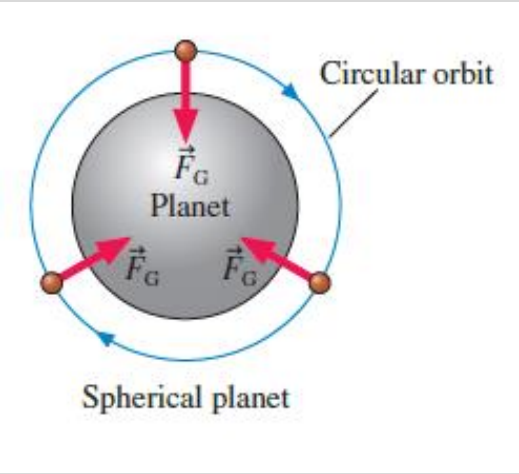
Circular orbit

- **Centripetal Acceleration** is the main reason to why **circular orbit** is possible.
- Recall the example of a satellite orbiting the Earth, notice that it is constantly being pulled toward the Earth by the **gravitational acceleration** g. If the Earth was flat, the satellite will crash.



Circular orbit

- However, since the Earth is approximately **spherical**, the satellite is actually under the pulling force in the direction toward the Center of the Earth.
- This acceleration is



$$a_r = \frac{v_{orbit}^2}{r} = g$$